HDOC - 64th day

FINDING THE DUPLICATE ELEMENT USING POINTERS

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Problem Statement

Write a C program that reads the size and elements of a variable-length integer array, nums[]. The array contains exactly one element repeated once. Pass the array to a function named duplicate_element() using its address. Use pointers to pass the array between the calling function and duplicate_element(). The function duplicate_element() must:

- Find the only repeated element in nums[] and display the repeated element.
- Find the result in a single iteration through the array by using an efficient approach such as the XOR technique.
- Assume that every other element appears exactly once.

Algorithm

1. Start.
2. Read the size of the array nums[].
3. Read the elements of nums[].
4. Pass nums and size to duplicate_element() using the array address.
5. Initialize result = 0.
6. Traverse the array and XOR every element with result.

7. XOR result with all integers from 1 to size - 1.

8. The remaining value in result is the repeated element.

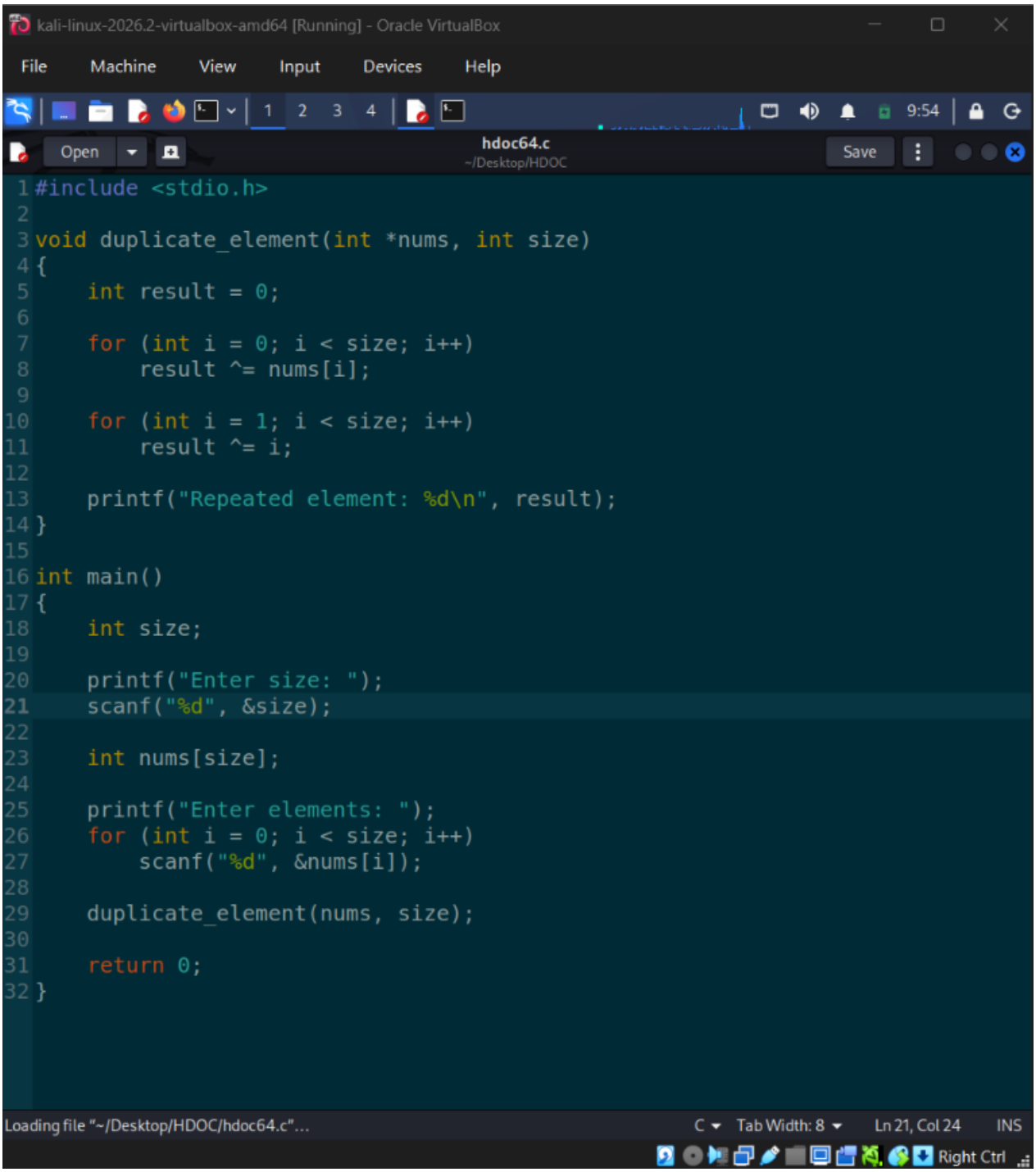
9. Display the repeated element.

10. Stop.

Test Case Table

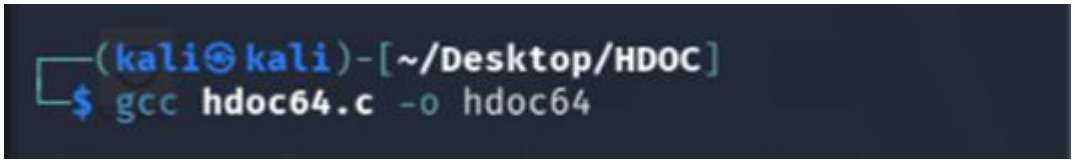
S. No.	Size	nums[]	Expected Output	Actual Output	Result
1	5	1 2 3 4 3	3	3	Pass
2	6	1 2 3 4 5 2	2	2	Pass
3	4	1 2 3 1	1	1	Pass
4	3	1 2 2	2	2	Pass
5	7	1 2 3 4 5 6 5	5	5	Pass

C Program


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Open
hdoc64.c
~/Desktop/HDOC
Save
#include <stdio.h>
void duplicate_element(int *nums, int size)
{
int result = 0;
for (int i = 0; i < size; i++)
result ^= nums[i];
for (int i = 1; i < size; i++)
result ^= i;
printf("Repeated element: %d\n", result);
}
int main()
{
int size;
printf("Enter size: ");
scanf("%d", &size);
int nums[size];
printf("Enter elements: ");
for (int i = 0; i < size; i++)
scanf("%d", &nums[i]);
duplicate_element(nums, size);
return 0;
}
Loading file "~/Desktop/HDOC/hdoc64.c"...
C
Tab Width: 8
Ln 21, Col 24
INS
Right Ctrl

Compilation and Execution

Compilation command: gcc hdoc64.c -o hdoc64



Execution command: ./hdoc64

Execution Screenshot:

